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International Journal of Project Management 33 (2015) 589 – 598

International Journal of Project Management

www.elsevier.com/locate/ijproman

Research on the phenomenon of asymmetric information in construction projects — The case of China

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Received 8 January 2014; received in revised form 5 October 2014; accepted 13 October 2014
Available online 11 November 2014

Abstract

From the perspective of information economics, asymmetric information leads to opportunistic behavior, which is the primary cause of loss and risk in the construction market. This research studies on the probability and impacts of asymmetric information phenomena systematically and focuses on information asymmetry among owners, contractors and supervisors. This is conducted by inviting construction industry participants in China to evaluate asymmetric information phenomena in project management by questionnaires. The evaluation includes two stages, i.e. bidding stage and performance stage. Risk significant index is used to identify the key asymmetric information phenomena. "Owner's financial condition" and "Owner's ability to pay progress payments" gains the highest score in these two stages respectively. The research helps to distinguish the primary and secondary order of asymmetric information phenomena and improves the efficiency of project management. Principal participants may use it as a reference to prevent and reduce the uncertainty and risk in project management.
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Keywords: Information asymmetry; Project management; Principal participants; China

1. Introduction

Project management is a complex system that faces many uncertainties. In the bidding stage, even though the bidder and the tenderer have a cooperative relationship in the construction of the project, the tenderer cannot know all about the benefit relationships between the bidders, so the tenderer will face the risk of accepting an irrational price which called adverse selection risk. For instance, in a tendering process which bidding without base price, some bidders privately reached "under the table agreement", consequently the tender price was higher than estimated cost more than 10% (Wang and Zhang, 2004). In performance stage, some contractors use inferior materials, reduce design standards and the necessary investment in construction or use other organizational measures to reduce the cost of construction project, which leads to substandard building products, namely the moral hazard. Such as a fishing port project in Hainan province, the contractors failed to follow the design drawings and construction standards in construction, which caused serious quality problems, and the 4000 million-invested fishing port became a "discard port", with a severe loss of state property.

From the information economics perspective, these phenomena are caused by Asymmetric Information (AI) among participants involved in the project. AI is put forward for the first time by Arrow (1963). Akerlof (1970) makes a further elaboration in the paper "lemon market", which creates the beginning of adverse selection. Then Spence (1973) proposes signal theory to solve the low efficiency of market adverse selection problems. Holmstrom (1979) makes an analysis of the moral risk and its observability, and points out that the individual's behavior is difficult to observe. Therefore, it leads to information asymmetry, which causes the presence of moral risk. In real economic activities, the participants may deliberately conceal part of the information for their own interests, so behavior agents are

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http://dx.doi.org/10.1016/j.ijproman.2014.10.007
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unaware of all the related knowledge and cannot foresee the future precisely when they make decisions, therefore they will face with uncertainty and risk inevitably (Sha et al., 2004; Xiang and Lu, 2011). Virtually uncertainty is a state of incomplete information, which can lead to opportunistic behavior, i.e. adverse selection and moral risk, is the primary cause of project risk. Therefore, the research on AIs in project management, not only has full feasibility, but also has practical significance. Summing up AIs in project management, eliminating the main AIs can effectively reduce the possibility of risks that participants may face with and increase the efficiency of project management (Qi and Sheng, 2008).

Meanwhile, the Chinese construction market plays an important role in global building market. According to the Top 225 International Contractors in 2012 released by ENR (Engineering News-Record), 41 companies in Chinese mainland are on the list, accounting for 18.22%. The total income of Chinese mainland contractors is 4383 billion U.S. dollars, accounting for 34.48% of the largest general contractors' general income (Chen and Li, 2012). Chinese construction market shares 18% in the global construction industry and grows steadily, so the research on Chinese project management information asymmetry has a significant practical significance, which can effectively promote the orderly and healthy development in global construction market.

This paper mainly researches on AIs in project management in China. Several AIs are collected from specialists and related literatures, ranked through questionnaires. Through the data analysis high probability and influential AIs in project management are summarized. Discussing about these AIs can help increase information transparency in the construction market, reduce the cost of information collection, and promote the healthy and orderly development of project management.

2. Literature review

In recent years, the researches on AI are widely involved in finance, insurance, labor, risk management and construction projects, etc. In this background, domestic and international relevant scholars have started in-depth study of the application of AIs in construction projects.

2.1. General situation of AIs in project management

One of the important applications of AI in project management is the principal–agent problem (Jager, 2008). Lampel et al. (1996) discuss about AI and the scientific technology innovation in large scale project. Zhang and Guan (2000) propose the information model of construction and AI problems, and quantity analysis method is used to study how to determine the moderate symmetry range. Then Li and Yu (2002) discuss the relationship among the information symmetry, engineering cost and engineering income, and do a theoretical discussion about moderate symmetry range between owners and contractors. To know well about AIs among the participants, Turner and Muller (2003) point out that the owner has a principal–agent relationship with the contractor. One of the key problems of the principal–agent relationship is "incentive". Therefore, how to explore the optimal incentive mechanism is the key to the success of the project. Verma and Terpenny (2005) put forward a dynamic project management thought based on the principal–agent model, which considers that the change of the principal participants' behavior can solve the problems of information asymmetry in project management. The principal–agent problems arise because of conflicting interests and AIs between principals and agents, and to improve the construction agent management model by establishing incentive and restraint mechanisms, strengthening contract management and implementing project guarantees and insurance system (Deng, 2006). Ceric (2012) uses the principal–agent theory to study the communication risk among the owner, the contractor and the project manager, he thinks that before the contract signing AI mainly exists between the owner and the contractor, and after signing the contract AI mainly exists between the owner and the project manager.

2.2. Mojar AIs in previous literatures

In recent years, the researches on AI are widely involved in finance, insurance, labor, risk management and construction projects, etc. In this background, domestic and international relevant scholars have started in-depth study of the application of AIs in construction projects. For instance, in the real estate market, developers are prone to take advantage of their own information to get high profits. Or they may not conform to the quality requirements or appear malicious speculation behavior in house sales (Cai and Jiang, 2003). Zhang and Guan (2000), Xiang et al. (2006), and Zheng (2010) emphatically discuss AIs between the owner and the contractor. In the bidding stage, owners are clearer about their own building intention, the financial condition, the project procedure, the transparency in evaluation of bid, etc. But the owner knows less about the contractor's technology, ability, equipment, management, and other information. In the contract implementation, the contractor knows more about their own building behavior than the owner, who is in the information advantage, but the contractor does not know about the credit and financial ability of the owner, at this point the contractor belongs to the information underdog. Huai and Wang (2004) focus on AIs between the owner and the supervisor. In the bidding stage, the supervisor's information advantages include the unit qualification, credit standing and financial position, performance, technology, the number of employees and their quality, etc. In the performance stage, the supervisor's information advantages mainly contain the number of supervision engineer, technical skills, professional moral risk, etc. Zhu and Li (2005) discuss about moral hazard of contractors under information asymmetry. In the contract implementation, the contractor may suits his own interests through a claim or improper means, such as cutting corners, and reduces the investment in the construction measures to reduce the actual cost. Ye and Zhang (2008) propose information advantage of the owner is construction intention and pay ability, which is information disadvantage of the supervisor and contractor. The qualification, credibility,

technology, quality and ability of the supervisor and contractor are information advantage for themselves, while these are information disadvantage for the owner. Wang (2013) shows AIs in the project management mainly from the perspective of contractors, including making the technical and business bid phase out, raising the material price standard, paraphrasing the high quota, etc.

2.3. Research on the prevention of adverse selection and moral hazard

From the perspective of information economics, domestic related researches show that the main cause of project risks is AI, and fundamentally to say, the project risk management is to solve the AIs which lead to the adverse selection and moral hazard problems (Schieg, 2008; Xiang and Xie, 2008). As to information asymmetry solutions, relevant scholars have also made a lot of researches. Wang and Zhang (2004) consider that in the bidding stage, to prevent adverse selection problems, tenderers can determine a reasonable bidding control price through the information screening, thus to eliminate malicious bidders, and then choose a reasonable low price as a successful bidder price after scientific scrutiny. According to Spence signal transfer theory, in the bidding stage, the bidders send a strong signal by advertising, providing engineering guarantee and establishing brand effect to the tenderers, so that the tenderers can make a proper identification (Liu and Hu, 2008). Fang (2013) points out that due to AI in the engineering construction bidding, which easily leads to integrity risk. To control the risk, the owner can promote commissioned bidding mode, provide fair trading platform, and strengthen information disclosure and information management. At the same time, in the performance stage, more attention should be paid to the default clause of the contract to ensure that the contractors are not willing to make "moral corruption" behavior even when they are pursuing maximum profits (Wei, 2009). The government should establish and improve relevant legal systems to format the evaluation system of the general contractor selection, in order to eliminate the opportunistic behaviors such as moral hazard caused by AI and incomplete contracts as much as possible (Li and Li, 2010).

An increasing number of scholars focus on asymmetric information in project management and much of them summarizes AIs among the owner, the contractor, the supervisor, etc. However, current literatures cannot make it clear that the possibility of AIs and their impact on project management. They did not rank AIs through quantitative research, which obstructs all participants from coping with high-probability and high-impact AIs specifically. Then they have to increase the cost of getting information, which reduces the potential returns. This paper is a further and quantitative study based on the analysis of Xiang et al. (2012). It makes a competitive research on the significant AIs in project management on the basis of the existing research, then summarizes high-probability and high-impact AIs and lays the foundation of researching the prevention and solution methods.

3. Research methodology

The research methodology selected for this research contains a comprehensive literature review and specialist interview, which were used to collect several common AI phenomena. AIs in this research were mainly sourced from Xiang et al. (2006), Ye and Zhang (2008), Xiang and Xie (2008), El-Sayegh (2008), Chan et al. (2011), Xiang et al. (2012), and so on, as was shown in Table 1. As information asymmetry is the root causes of construction project risk (Garmaise and Moskowitz, 2004; Xiang and Lu, 2011), we summarized some potential risk factors in previous literatures equal to AIs in the research. To the best of the authors' knowledge, construction project management were put into two categories, i.e. bidding stage and performance stage, including information asymmetry phenomena among owners, contractors and supervisors. The owner, also known as the development organization, is the buyer of the construction project products, whom will be called the principal. The contractor is called the agent and is the seller of the construction project products. The supervisor is an independent third party, and is an employee of the development organization in project management. There is a principal–agent relationship between the owner and the contractor established by the construction contract, while there is a principal–agent relationship between the owner and the supervisor defined by the supervision contract.

After analyzing and selecting from those articles, 26 AIs were set out in the questionnaire. In order to develop a comprehensive list of AIs, 5 specialists were asked to test whether the questions were intelligible, unambiguous, etc. So the original AI list was sent to the specialists by email separately. As a result, valuable feedbacks were obtained to improve the quality of the questionnaire and 4 extra AIs were added, including design process and depth of the construction project, the differences of the project quality requirements between owners and specifications, contractor's ability to deal with major emergencies in the performance stage and supervisor's ability to control the contractor. Therefore 30 AIs were obtained in total, which is shown in Table 1. Then network questionnaires were given out to the construction industry practitioners to research the possibility of these phenomena and their impact level on project objectives. The questionnaire consisted of three sections. Section 1 collects general information about the respondents. Section 2 selects information about projects they have participated in. Section 3 carries a total of 30 AIs associated with construction projects and asks respondents to review and indicate the possibility of these AIs and the level of impact on project management. The possibility is divided into 5 degrees, where 1 = extremely small, 2 = small, 3 =medium, 4 = large, and 5 = extremely large. The impact is divided into 5 degrees as well, where 1 = extremely unimportant, 2 = unimportant, 3 = medium, 4 = important, and 5 = extremely important. The complete questionnaire is added to the appendix.

Table 1
AIs list from related literatures and specialists.

AIs	Advantage	Disadvantage	Source
Bidding stage			
A1 Owner's business reputation	Owner	Contractor/supervisor	Xiang and Xie (2008)
A2 Owner's detailed intention of the construction project	Owner	Contractor/supervisor	Zhang and Guan (2000)
A3 Complete formalities of the construction project	Owner	Contractor/supervisor	Xiang et al. (2006)
A4 Owner's requirement in design depth of the construction project	Owner	Contractor/supervisor	Specialists
A5 The transparency in bid evaluation	Owner	Contractor/supervisor	Wei (2009)
A6 Owner's financial condition	Owner	Contractor/supervisor	Xiang et al. (2006), Xiang et al. (2012)
A7 Difference between specification requirements and the owner's understanding	Owner	Contractor/supervisor	Specialists
B1 Contractor's qualification and achievement	Contractor	Owner	Xiang et al (2006)
B2 Contractor's business reputation	Contractor	Owner	Xiang et al. (2006)
B3 Contractor 's technical and management capabilities	Contractor	Owner	Zhang and Guan (2000)
B4 Validity of the tender documents	Contractor	Owner	Wang (2013)
C1 Supervisor's business reputation	Supervisor	Owner	Huai &Wang (2004) Xiang et al. (2012)
C2 Supervisor's qualification and achievement	Supervisor	Owner	Huai &Wang (2004) Xiang et al. (2012)
C3 Supervisor's technical strength and staff quality	Supervisor	Owner	Huai &Wang (2004) Xiang et al. (2012)
Performance stage			
D1 Owner's change in scope of work	Owner	Contractor/Supervisor	Chan et al. (2011)
D2 Owner's change of the design	Owner	Contractor/supervisor	El-Sayegh (2008)
D3 Owner's supply of the raw materials and equipment	Owner	Contractor/supervisor	Mu and Mu (2006)
D4 Owner's ability to pay progress payments	Owner	Contractor/supervisor	Fang et al. (2004)
D5 Owner's control of the breach of contract	Owner	Contractor/supervisor	Fang et al. (2004)
E1 Contractor's understanding of the construction intent	Contractor	Owner	Xiang et al. (2006),
E2 Contractor's understanding on construction conditions	Contractor	Owner	Zhan and Liu (2007)
E3 Contractor's ability to control construction progress	Contractor	Owner	Xiang et al. (2012)
E4 Contractor's construction methods and technology	Contractor	Owner	Xiang et al. (2006)
E5 Contractor's ability to deal with major emergencies	Contractor	Owner	Specialists
E6 Contractor's claims behavior in construction process	Contractor	Owner	Zhu &Li (2005)
F1 Supervisor's understanding of the situation at the scene	Supervisor	Owner	Xiang et al. (2012)
F2 Supervisor's supervision of the construction process	Supervisor	Owner	Xiang et al. (2012)
F3 Supervisor's effort to fulfill contractual duties	Supervisor	Owner	Xiang et al. (2012)
F4 Supervisor's collusion with contractors	Supervisor	Owner	Huai &Wang (2004)
F5 Supervisor's ability to control the contractor	Supervisor	Owner	Specialists

4. Data collection

After a small refinement, the research team distributed the questionnaires on a website (www.sojump.com). Then the links were sent to related participants in project management. All respondents were contacted beforehand to make sure that they were willing to join this research. The analysis results in this research were based on the data collected from a network questionnaire, which is one of the most cost effectively ways to involve a large number of people in the process in order to achieve better results.

4.1. Sample size

After a two-week waiting period, 22 feedbacks were received. Then the questionnaire was distributed in text form, and 150 were given out to professional related participators in the construction field. Then 100 feedbacks were collected, of these 17 were completely empty records and 12 were mostly incomplete. All of these were identified as invalid and could not be included in the data analysis since it would affect the classification and analysis results. Therefore the total valid questionnaires were 93, including the questionnaires collected from the website. This represented a valid response rate of 54.07%, which is acceptable according to Moser and Kalton's assertion (1971).

4.2. Sample composition

The respondents are all industry practitioners, their organization types including government sector, designing institute, construction organization, development organization, supervision unions, intermediary companies, research institutions and so on. The details of the distribution (in percentage) about respondents are shown in Table 2. They have an average of 10 years' work experience in the construction field, and it is evident that more than 80% of respondents have worked more than 5 years in this field. As to the respondents' occupations at work, almost half of them are project managers. This research also consists of the basic situation of the projects that the respondents have participated in, as shown in Table 3. As we can see, nearly 90% are civil and industrial constructions. More than 90% of the bidding types are open tendering and selected bidding. As to the investment entities of the projects, we can

Table 2
Personal situation about the respondents.

Category	Respondents	
	Number	%
Type of organization		
Government sector	4	4.3
Design institute	2	2.15
Construction organization (i.e. Contractor)	62	66.67
Development organization (i.e. Owner)	17	18.28
Supervision company (i.e. Supervisor)	5	5.38
Intermediary companies	1	1.08
Research institution	1	1.08
Others	1	1.08
Years of working		
Less than 2 years	4	4.3
2–5 years	13	13.98
5–10 years	14	15.05
10–15 years	23	24.73
More than 15 years	39	41.94
Type of position		
Company controller	6	6.45
Corporate executive	10	10.75
Division principle	13	13.98
Project management personnel	46	49.46
Project specific participant	15	16.13
Others	3	3.23

Table 3
Basic situation of the project.

Category	Respondents	
	Number	%
Item type		
Civil and industrial construction	80	86.02
Traffic and road project	1	1.08
Industrial construction	10	10.75
Water port project	0	0
Others	2	2.15
Bidding type		
Open tendering	54	58.06
Selected bidding	30	32.26
Negotiable bidding	7	7.53
Others	2	2.15
Project process		
Construction in process	74	79.57
Have been completed	15	16.13
Not started	4	4.3
Investment		
Corporate (private) investment	59	63.44
Government investment	25	26.88
Government and private capital combined	8	8.6
Others	1	1.08
Financing mode		
Self-financing	52	55.91
Bank loans	21	22.58
BT	8	8.6
BOT	3	3.23
PPP	3	3.23
Others	6	6.45
Contracting form		
Unit price contract	36	38.71
Lump-sum contract	24	25.81
Cost-plus-fee contract	1	1.08
Reference to the local fixed rate to fall	29	31.18
Others	3	3.23

see that nearly 65% are private investment. Project financing modes are reflected in this table too, and almost 60% are self-financing. Contracting forms have shown that most of them are unit price contracts and forms of reference to the local fixed rate to fall and lump-sum contracts are also used very often.

5. Data analysis

5.1. Analysis method

The research feedback includes two groups of data, the possibility of each AI and its level of impact on project management. The risk significant index developed by Shen et al (2001) is considered effective in previous works (Fang et.al., 2004; Zou et al., 2006; Zou et al., 2007), so it is used for this research as well.

With respect to the impact on project management, the significance score for each AI assessed by each respondent can be calculated through Eq. (1).

$$S_i = P_i * D_i \tag{1}$$

where S_i = significance score of AI i. P_i = probability of the occurrence of AI i and D_i = degree of impact of AI i.

Then normalization process has been done to identify key AIs, this process can be described as Eq. (2).

$$S_i' = S_i/(5 * 5) = S_i/25 \tag{2}$$

5.2. Analysis

Through the overall research results in Table 4, it can be seen that more than half of the respondents identify that in China the degree of AIs in project management is high. When it concerns to degree of the lost and influence caused by AIs, almost 63% of the respondents agree that it is large.

As to the specific phenomenon research, the results of the scores and ranking are shown in Table 5. Please note that the method for calculating the significance score may overlook those AIs with a less possibility but a high level of impact on project management, which should be taken into account in the project management practice and which is not the focus of this research. SPSS18.0 is used to test the reliability of the data. Individual reliability of variables is accepted to be satisfactory if the Cronbach's alpha coefficients reported for each item are greater than the threshold value of 0.7 (Eybpoosh et al., 2011). Table 5 demonstrates that all scales are individually reliable, as the alpha coefficients reported for each item range from 0.821 to 0.913.

Table 4
The overall research result.

Category	Respondents	
	Number	%
The possibility of asymmetry information		
Extremely small	1	1.08
Small	6	6.45
Medium	24	25.81
Large	57	61.29
Extremely large	5	5.38
The degree of the impact		
Extremely unimportant	1	1.08
Unimportant	4	4.3
Medium	24	25.81
Important	58	62.37
Extremely important	6	6.45

All AIs observed in the questionnaire can happen to any construction projects. The main purpose of this research is to ascertain the key AIs that can significantly influence the delivery of construction projects. Hence, only the top 10 ranked ones are chosen as key AIs in line with other similar research (Tam et al., 2004). The top 10 AIs in this research are shown in Table 6, including 5 AIs in the bidding stage and 5 AIs in the performance stage.

6. Discussions

By ranking the AIs in these two stages we can conclude factors that are highlighted in project management. There is principal–agent relationship between owners and contractors, and owners and supervisors. In such a relationship, each participant is in pursuit of its own interest, it may hide some information which results in AIs. As these AIs are all project participants orientated, how to effectively get different participants to manage them in the process is decisive to the project success. The abbreviation of the top 10 AIs can be seen in Table 6. Table 6 presents that the AIs in the bidding stage are mostly related to the owner and the contractor. While in the performance stage, AIs exist in the owner more frequently, as we can see that 4 of the top 5 AIs related to the owner.

As to the questionnaire analysis, in the bidding stage, there is still existing information blind area between the owner and the contractor. The owner's poor financial condition can lead to contractor's underwritten, which does harm to the contractor's economic situation. Some contractors overstate their construction technology in the bidding stage, in the end the quality of the construction project is not up to the standards. The supervisor has the opportunity to reduce the real technical force and staff quality after winning the bid, which results in hidden problems in project

Table 5
The scores and rank of the AIs.

AIs	The possibility of AIs	The impacts of AIs	Significance scores (after normalization)	Rank	Cronbach's alpha
Bidding stage	**mean**	**mean**			
A1 Owner's business reputation	3.23	3.57	0.46	7	
A2 Owner's detailed intention of construction project	3.22	3.19	0.41	9	
A3 Complete formalities of the construction project	2.89	2.91	0.34	11	
A4 Owner's requirement in design depth of the construction project	3.28	3.49	0.46	8	0.831
A5 The transparency in bid evaluation	3.61	3.57	0.52	4	
A6 Owner's financial condition	3.96	4.14	0.66	1	
A7 Difference between specification requirements and the owner's understanding	2.85	3.03	0.35	10	
B1 Contractor's qualification and achievement	2.68	3.05	0.33	12	
B2 Contractor's business reputation	3.3	3.55	0.47	6	0.890
B3 Contractor's technical and management capabilities	3.54	3.87	0.55	2	
B4 Validity of the tender documents	3.47	3.7	0.51	5	
C1 Supervisor's business reputation	2.8	2.43	0.27	13	
C2 Supervisor's qualification and achievement	2.65	2.57	0.27	14	0.821
C3 Supervisor's technical strength and staff quality	3.62	3.59	0.52	3	
Performance stage					
D1 Owner's change in scope of work	3.46	3.56	0.49	4	
D2 Owner's change of the design	3.48	3.66	0.51	2	
D3 Owner's supply of the raw materials and equipment	3.29	3.43	0.45	8	0.897
D4 Owner's ability to pay progress payments	3.85	4.14	0.64	1	
D5 Owner's control of the breach of contract	3.4	3.74	0.51	3	
E1 Contractor's understanding of the design intent	2.83	3.29	0.37	12	
E2 Contractor's understanding on construction conditions	2.84	3.23	0.37	13	
E3 Contractor's ability to control construction progress	3.23	3.6	0.41	9	0.891
E4 Contractor's construction methods and technology	3.03	3.37	0.47	5	
E5 Contractor's ability to deal with major emergencies	3.24	3.54	0.46	6	
E6 Contractor's claims behavior in construction process	3.35	3.4	0.46	7	
F1 Supervisor's understanding of the situation at the scene	2.6	2.78	0.29	16	
F2 Supervisor's supervision of the construction process	2.96	3.2	0.38	11	
F3 Supervisor's effort to fulfill contractual duties	2.75	2.99	0.33	15	0.913
F4 Supervisor's collusion with contractors	2.99	3.38	0.4	10	
F5 Supervisor's ability to control the contractor	2.87	3.03	0.35	14	

Table 6
Top 10 AIs.

Project stage	Rank	AIs
Bidding stage	1	A6 Owner's financial condition
	2	B3 Contractor 's technical and management capabilities
	3	C3 Supervisor's technical strength and staff quality
	4	A5 The transparency in bid evaluation
	5	B4 Validity of the tender documents
Performance stage	1	D4 Owner's ability to pay progress payments
	2	D2 Owner's change of the design
	3	D5 Owner's control of the breach of contract
	4	D1 Owner's change in scope of work
	5	E4 Contractor's construction methods and technology

supervision. Adverse selection between them is due to the obstruction of information transmission. In the contract implementation, the owner has more information advantage than contractor in finance, contract or design changes, which seriously affects the contractor's construction schedule. Some local governments and departments hastily build a number of projects without enough funds, then resulting in arrears. On the other hand, the contractor's construction methods and technology are difficult to be observed by the owner. Then some contractors who are winning at low prices cut corners and sacrifice the project quality for their own interests. So the core asymmetric information phenomenon in the performance stage is moral hazard. According to the survey results, the research team provides the following recommendations.

(1) In the bidding stage, the major participants, i.e. the owner, the contractor, and the supervisor, should share more relevant information with each other, and work cooperatively to diminish AIs. Project management information platform can be built to eliminate information transfer barriers, improve the information flow, realize information sharing, and reduce the information asymmetry. That is to say, participants can precaution asymmetric information through an information publication platform.
(2) In the performance stage, taking advantage of the information superiority side and building corresponding restraint mechanism of the participant's behavior can prevent moral hazard. As to the progress payment, arrears of payment may be restrained by setting up two-way guarantee mechanism between the owner and the contractor. Inevitable changes in construction should be conveyed timely to avoid information obstruction. New project management patterns, such as partnering management, project dynamic alliance, and the supply chain management can effectively change the confrontation relationship between the owner and the contractor, and strengthen the competitiveness of the whole cooperation organization through the information exchange and sharing.

7. Conclusions

Asymmetric information phenomena in construction projects can lead to the risk of moral hazard and adverse selection, which is harmful to the project management. As the time of big data is coming, considering the complex information of construction management, risk management should pay more attention to information management among participants. So it is essential to know well about which phenomenon may occur frequently, then we can take measures to control them. This research endeavors to identify key AIs associated with the achievement of project objectives. On the basis of a research with industry practitioners who own robust experience and knowledge of construction projects, top 5 key AIs are highlighted on a comprehensive assessment of their possibility and impacts degree on project management in each stage. In the bidding stage, owner's financial condition is the significant AI, while in the performance stage owner's ability to pay progress payments takes the first place. Each project participant should take prior control and prevent these AIs in reducing the loss of information asymmetry in project management as well.

Conflict of interest

The authors declared that they have no conflicts of interest to this work.

Acknowledgments

The work described in this paper was fully supported by a joint grant from Chongqing Social Science Foundation Project (2013ZDZZ02) and Project No. CDJKXB13004 and No. 106112012CDJSK5503 supported by the Fundamental Research Funds for the Central Universities. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors.

Appendix A. The questionnaire about information asymmetry in project management

Part I: Basic Situation

1. The type of your organization:
   - ☐ Government sector
   - ☐ Design institute
   - ☐ Construction organization
   - ☐ Development organization
   - ☐ Supervision company
   - ☐ Intermediary company

☐ Research institution
☐ Others

2. How long have you participated in work:

☐ Less than 2 years
☐ 2–5 years
☐ 5–10 years
☐ 10–15 years
☐ More than 15 years

3. The type of your situation:

☐ Company controller
☐ Corporate executive
☐ Division Principle
☐ Project management personnel
☐ Project Specific Participant
☐ Others

Part 2: the basic situation of the projects you participate in

4. Project provinces (municipalities): _____
5. Item type:

☐ Civil and industrial construction
☐ Traffic and road project
☐ Industrial construction
☐ Water port project
☐ Others

6. Bidding type:

☐ Open tendering
☐ Selected bidding
☐ Negotiable bidding
☐ Others

7. Project process:

☐ Construction in progress
☐ Have been completed
☐ Not started

8. Project investment entities:

☐ Corporate (private) investment
☐ Government investment
☐ Government and private capital combined
☐ Others

9. Project financing mode:

☐ Self-financing
☐ Bank loans
☐ BT
☐ BOT
☐ PPP
☐ Others

10 Project contract form:

☐ Unit price contract
☐ Lump-sum contract
☐ Cost-plus-fee contract
☐ Reference to the local fixed rate to fall
☐ Others

Part 3: information asymmetry research

The so-called asymmetric information is about asymmetric distribution of information between mutually corresponding master. People who have adequate information often stay in a relatively favorable position, while the information-poor people stay in a disadvantageous position. In project management, information asymmetries are also common.

11. The overall impression research
 1) Do you think that the possibility of asymmetric information in project management is common ().
    ☐ Extremely small
    ☐ Small
    ☐ Medium
    ☐ Large
    ☐ Extremely large
 2) The impact on project management caused by information asymmetry ().
    ☐ Extremely unimportant
    ☐ Unimportant
    ☐ Medium
    ☐ Important
    ☐ Extremely important
12. The specific phenomenon research

This section carries a total of 30 AIs associated with construction projects and asks respondents to review and indicate the possibility of these AIs and the level of impact on project management. The possibility is divided into 5 degrees, where 1 =extremely small, 2 = small, 3 = medium, 4 = large, and 5 = extremely large. The impact is divided into 5 degrees as well, where 1 = extremely unimportant, 2 = unimportant, 3 = medium, 4 = important, and 5 = extremely important. Please select the appropriate option in your judgment.

Asymmetric Information Phenomena		**Asymmetry**		**Possibility**	**Total Impact**
Participants	**Bidding Stage**	**Advantage**	**Disadvantage**	**1 ~ 5**	**1 ~ 5**
Owner	**A1** Business reputation	**Owner**	**Contractor/Supervisor**		
	A2 Detailed intention of the construction project	**Owner**	**Contractor/Supervisor**		
	A3 The formality process of the construction project	**Owner**	**Contractor/Supervisor**		
	A4 Design process and depth of the construction project	**Owner**	**Contractor/Supervisor**		
	A5 The transparency in evaluation of bid	**Owner**	**Contractor/Supervisor**		
	A6 Financial condition	**Owner**	**Contractor/Supervisor**		
	A7 The differences of the project quality requirements between owners and specifications	**Owner**	**Contractor/Supervisor**		
Contractor	**B1** The qualification and achievement	**Contractor**	**Owner**		
	B2 The business reputation	**Contractor**	**Owner**		
	B3 The technical and management capabilities	**Contractor**	**Owner**		
	B4 Validity of the tender documents	**Contractor**	**Owner**		
Supervisor	**C1** The business reputation	**Supervisor**	**Owner**		
	C2 The qualification and achievement	**Supervisor**	**Owner**		
	C3 The technical strength and staff quality	**Supervisor**	**Owner**		
Participants	**Contract Implementation**	**Advantage**	**Disadvantage**	**Possibility 1 ~ 5**	**Total Impact 1 ~ 5**
Owner	**D1** Change the goal of the construction project	**Owner**	**Contractor/Supervisor**		
	D2 Change the design	**Owner**	**Contractor/Supervisor**		
	D3 Supply of the raw materials and equipment	**Owner**	**Contractor/Supervisor**		
	D4 Ability to pay progress payments	**Owner**	**Contractor/Supervisor**		
	D5 Control of the breach of contract	**Owner**	**Contractor/Supervisor**		
Contractor	**E1** Understanding of the design intent	**Contractor**	**Owner**		
	E2 Understanding on construction conditions	**Contractor**	**Owner**		
	E3 Ability to control construction process	**Contractor**	**Owner**		
	E4 Construction methods and technology	**Contractor**	**Owner**		
	E5 Ability to deal with major emergencies	**Contractor**	**Owner**		
	E6 Claims behavior	**Contractor**	**Owner**		
Supervisor	**F1** Understanding of the situation at the scene	**Supervisor**	**Owner**		
	F2 Supervision of the construction process	**Supervisor**	**Owner**		
	F3 Effort to fulfill contractual duty	**Supervisor**	**Owner**		
	F4 Collusion with contractors	**Supervisor**	**Owner**		
	F5 Ability to control the contractor	**Supervisor**	**Owner**		

References

Akerlof, G., 1970. The market for "lemons" quality uncertainty and market mechanism. Q. J. Econ. 84 (3), 488–500.

Arrow, K.J., 1963. Uncertainty and the welfare economics of medical care. Am. Econ. Rev. 53 (5), 941–973.

Cai, N., Jiang, J.N., 2003. The application of asymmetric information theory in the real estate market. Value Eng. (01), 19–21.

Ceric, A., 2012. Strategies for minimizing information asymmetries in construction projects: project managers' perceptions. Working Paper Proceedings, Engineering Projects Organizations Conference, Rheden, Nizozemska, July 10-12.

Chan, D.W., Chan, A.P., Lam, P.T., Yeung, J.F., Chan, J.H., 2011. Risk ranking and analysis in target cost contracts: empirical evidence from the construction industry. Int. J. Proj. Manag. 29 (6), 751–763.

Chen, S.Y., Li, R.H., 2012. 2012 ENR225 largest contractors double list released. Chin. Invest. Des. (9), 48–50.

Deng, Z.M., 2006. Construction-agent system project management pattern study based on principle-agent theory. J. Chongqing Jiaotong Univ. (02), 128–131.

El-Sayegh, S.M., 2008. Risk assessment and allocation in the UAE construction industry. Int. J. Proj. Manag. 26 (4), 431–438.

Eybpoosh, M., Dikmen, I., Talat Birgonul, M., 2011. Identification of risk paths in international construction projects using structural equation modeling. J. Constr. Eng. Manag. 137 (12), 1164–1175.

Fang, Y.H., 2013. On the clean and honest administration risk and its prevention and control about tendering and bidding in the domain of project construction. J. Anhui Univ. Technol. (Social Sciences) (01), 34–35.

Fang, D.P., Zhu, D.F., Fong, Patrick, S.W., Shen, L.Y., 2004. Risks in Chinese construction market—contractors' perspective. J. Constr. Eng. Manag. 130 (6), 853–861.

Garmaise, M.J., Moskowitz, T.J., 2004. Confronting information asymmetries: evidence from real estate markets. Rev. Financ. Stud. 17 (2), 405–437.

Holmstrom, B., 1979. Moral hazard and observability. Bell J. Econ. 10 (1), 74–91.

Huai, X.F., Wang, X.L., 2004. Analysis of information asymmetry between client and supervision in construction engineering project. Constr. Manag. Modernization (05), 45–47.

Jäger, C., 2008. The Principal–Agent Theory within the Context of Economic Sciences. Herstellung und Verlag, Books on Demand GmbH, Norderstadt.

Lampel, J., Miller, R., Floricel, S., 1996. Information asymmetries and technological innovation in large engineering construction projects. RandD Manag. 26 (4), 357–369.

Li, Z.L., Li, S.D., 2010. Research on the mode of EPC based on principal–agent theory. Constr. Econ. (06), 69–71.

Li, B.Y., Yu, L., 2002. Research of information asymmetry in construction market. J. Ind. Technol. Econ. (04), 87–88.

Liu, G.W., Hu, G.Q., 2008. Asymmetric information and construction projects. Product. Res. (02) (64-65 + 109).

Moser, C.A., Kalton, G., 1971. Survey methods in social investigation. Survey methods in social investigation2, p. 549.

Mu, M., Mu, J., 2006. Risk management on EPC project. Int. Econ. Cooperation (5), 40–43.

Qi, A.B., Sheng, F., 2008. Research of project risk management facing the uncertainty. Theor. Invest. (02), 70–73.

Schieg, M., 2008. Strategies for avoiding asymmetric information in construction project management. J. Bus. Econ. Manag. 9 (1), 47–51.

Sha, K.X., Song, T., Zhao, J.K., Yin, T., 2004. Rectifying and standardizing the construction market from asymmetric information perspective. Constr. Econ. (1), 82–85.

Shen, L.Y., Wu, G.W., Ng, C.S., 2001. Risk assessment for construction joint ventures in China. J. Constr. Eng. Manag. 127 (1), 76–81.

Spence, M., 1973. Job market signaling. Q. J. Econ. 87 (3), 355–374.

Tam, C.M., Zeng, S.X., Deng, Z.M., 2004. Identifying elements of poor construction. Saf. Manag. Chin. (42), 569–586.

Turner, R., Muller, R., 2003. On the nature of the project as a temporary organization. Int. J. Proj. Manag. 21 (1), 1–8.

Verma, A., Terpenny, J., 2005. Dynamic project management: a principal–agent based approach. Int. J. Phys. Distrib. Logist. Manag. 36 (7), 490–506.

Wang, W.Q., 2013. The analysis of asymmetric information and the internal control measures in project management. Contemp. Econ. (05), 60–62.

Wang, L., Zhang, C.F., 2004. Information asymmetry risk in the bidding process of construction project. Econ. Tribune (09), 149–153.

Wei, X.M., 2009. Project construction based on the theory of asymmetric information. CO-Operative Econ. Sci. (22), 78–79.

Xiang, P.C., Lu, T., 2011. Investigating generating mechanism of project risk. Constr. Econ. (09), 43–45.

Xiang, P.C., Xie, L.L., 2008. Project risk management based on the theory of asymmetric information. Constr. Econ. (01), 59–62.

Xiang, P.C., Ren, H., Guo, F., 2006. Asymmetric information theory and its application in engineering project management. J. Chongqing Jianzhu Univ. (01), 119–122.

Xiang, P.C., Zhou, J., Zhou, X.Y., Ye, K.H., 2012. Construction project risk management based on the view of asymmetric information. J. Constr. Eng. Manag. 138 (11), 1303–1311.

Ye, C.Q., Zhang, X., 2008. Analysis of the problem of asymmetric information in project management. Proj. Manag. (06) (53-55 + 64).

Zhan, J.C., Liu, J.Y., 2007. The risk caused by invalid communication and countermeasures. Int. Econ. Cooperation (11), 53–56.

Zhang, D.Q., Guan, K., 2000. Study on information model of construction industry and asymmetric information. J. Harbin Univ. Civil Eng. Archit. (04), 93–95.

Zheng, X.C., 2010. The application of asymmetric information theory in engineering project management. Sci. Technol. West China (32) (49-50 + 44).

Zhu, B., Li, Q.M., 2005. Analysis of the moral risk model during construction process under asymmetric information. J. Chongqing Jianzhu Univ. (04) (102-105 + 110).

Zou, P.X., Zhang, G.M., Wang, J.Y., 2006. Identifying key risks in construction projects: life cycle and stakeholder perspectives. Pacific Rim Real Estate Society Conference (January).

Zou, P.X., Zhang, G., Wang, J., 2007. Understanding the key risks in construction projects in China. Int. J. Proj. Manag. 25 (6), 601–614.